Exercise 1 SQL FUNDAMENTALS : SELECT & FILTERING

01 Selecting & Sorting

employees

id	first_name	last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

1. SELECT *
FROM 'workspace'.'default'.'employees' ;

id	first_name	last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. SELECT DISTINCT department
FROM 'workspace'.'default'.'employees' ;

department
IT
HR
Finance
Marketing
IT

3. SELECT first_name,
last_name,
Salary
FROM 'workspace'.'default'.'employees'
ORDER BY Salary DESC ;

first_name	last_name	Salary
Eva	Naidoo	72 000
Alice	Green	70 000
Cathy	Zulu	65 000
David	Moroena	50 000
Brian	Lee	45 000

4. SELECT id,
first_name,
last_name,
Salary AS highest_paid
FROM 'workspace'.'default'.'employees'
ORDER BY Salary DESC
LIMIT 3;

id	first_name	last_name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02 FILTERING with WHERE

05. SELECT id, first_name, last_name, department
FROM 'workspace'.'default'.'employees'
WHERE department = 'IT';

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SELECT id, first_name, last_name, department, Salary
FROM 'workspace'.'default'.'employees'
WHERE department = 'Finance'
AND Salary > 60000 ;

id	first_name	last_name	department	Salary
3	Cathy	Zulu	Finance	65000

7. SELECT id, first_name, last_name, department
FROM 'workspace'.'default'.'employees'
WHERE department = 'HR' OR department = 'Marketing' ;

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SELECT id, first_name, last_name, department
FROM 'workspace'.'default'.'employees'
WHERE department NOT IN ('IT') ;

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. SELECT id, first_name, last_name, department
FROM 'workspace'.'default'.'employees'
WHERE department IN ('IT', 'HR', 'Finance') ;

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT id, first_name, last_name, department, salary, city
FROM 'workspace'.'default'.'employees'
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg' ;

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg